

Vancouver Community Health Resource Priority

Community vulnerability, census uncertainty and mapped healthcare supply gaps

Independent portfolio analysis using public/open data. Priority scores support comparative planning analysis and do not represent validated healthcare need.

CHSA Name

CHSA Name

All



Total Population

663.9K

Total CHSAs

19

Total Facilities

153

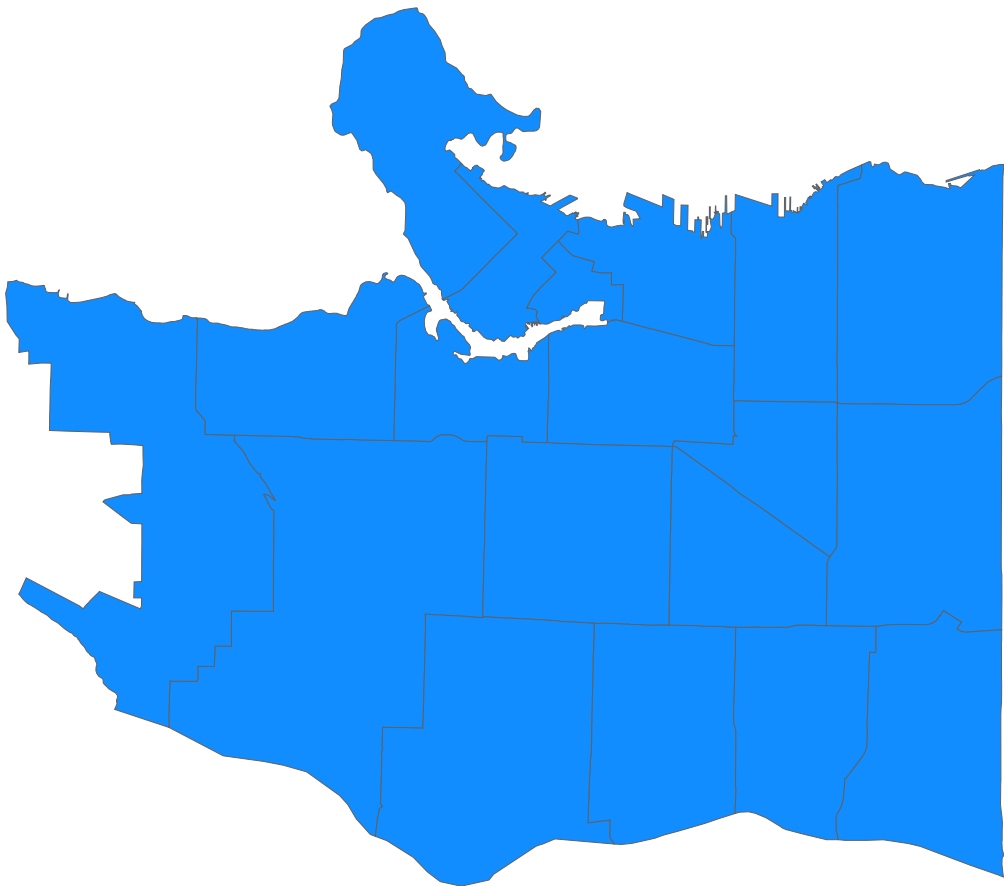
Facilities per 10K Population

2.30

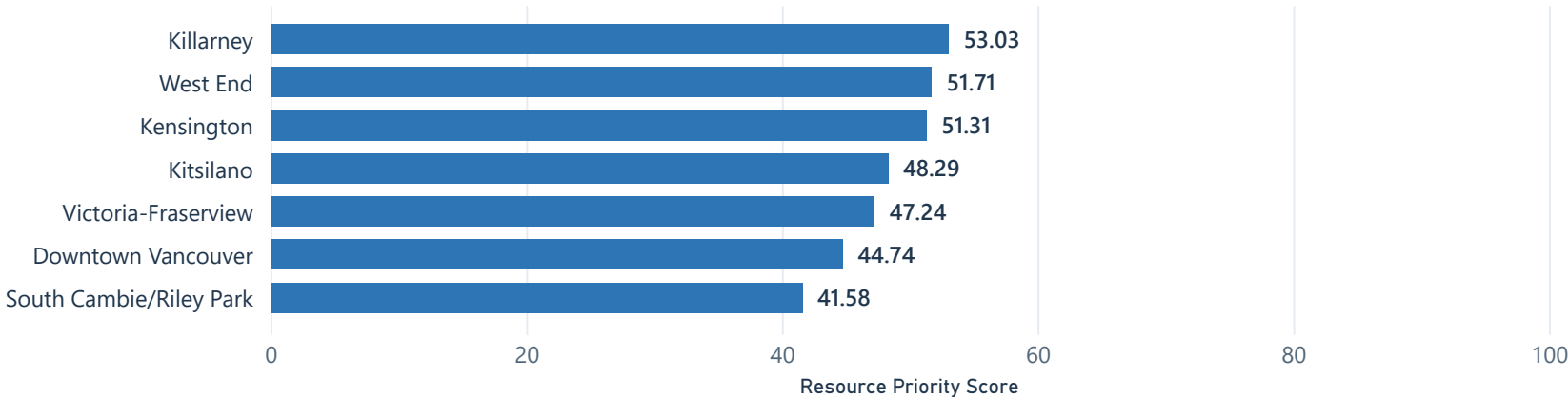
Weighted GNR

6.51%

Where Resource Priority Is Highest

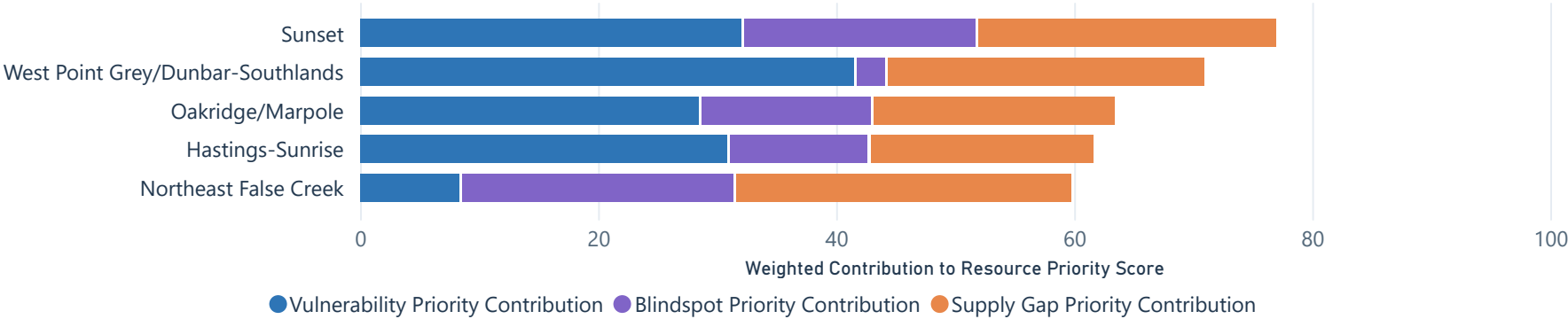


Highest-Priority Communities



What Drives Priority?

Balanced weights: Vulnerability 45% · Data Blindspot 25% · Supply Gap 30%



Management Insight

Sunset shows the strongest multi-component priority signal and remains first or second across all tested weighting scenarios. West Point Grey/Dunbar-Southlands is driven strongly by the age-structure vulnerability proxy and relative mapped-facility scarcity, while Northeast False Creek is influenced primarily by census uncertainty and zero mapped ODHF facilities.

Methodological caution

Mapped facility locations are a supply-location proxy, not a measure of capacity, staffing, utilization, accessibility or VCH ownership.

Community Vulnerability

Where age-structure vulnerability is highest across Vancouver communities

Age structure is used as a comparative vulnerability proxy and does not represent validated healthcare need.

CHSA Name

CHSA Name

All

Total Population

663.9K

Youth %

11.4%

Senior %

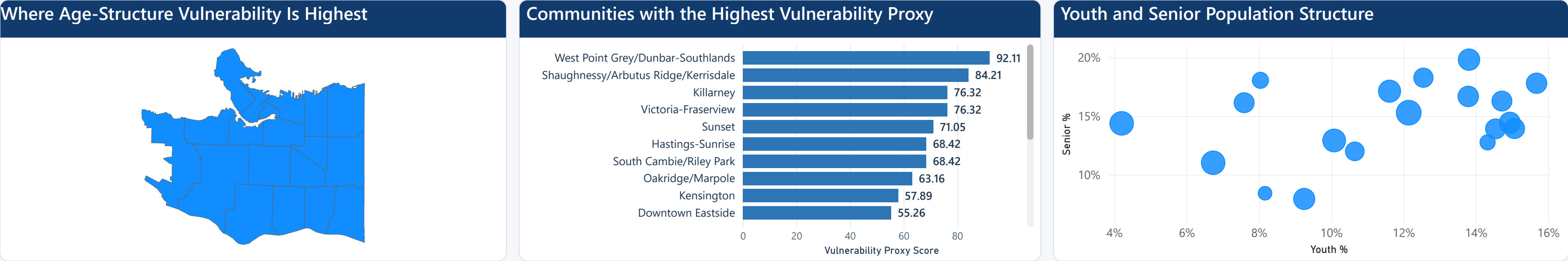
14.8%

Vulnerability Proxy (Avg./Selected)

52.63

Total CHSAs

19



Age Structure of Higher-Vulnerability Communities

Youth and senior shares shown; working-age share not displayed.

West Point Grey/Dunbar-Southlands	15.7%	17.8%
Shaughnessy/Arbutus Ridge/Kerrisdale	13.8%	19.8%
Killarney	14.7%	16.3%

Community Vulnerability Detail

CHSA Name	Total Population	Youth %	Senior %	Vulnerability Proxy Score	Resource Priority Score	Priority Rank
West Point Grey/Dunbar-Southlands	34866.0	15.7%	17.8%	92.11	70.92	2
Shaughnessy/Arbutus Ridge/Kerrisdale	38640.0	13.8%	19.8%	84.21	56.71	7
Killarney	33258.0	14.7%	16.3%	76.32	53.03	8
Victoria-Fraserview	30753.0	12.6%	18.3%	76.32	47.24	13
Total	663894.0	11.4%	14.8%	52.63	51.05	1

Management Insight

West Point Grey/Dunbar-Southlands has the strongest age-structure vulnerability proxy (92.11), combining the highest youth share (15.68%) with a high senior share (17.80%). Shaughnessy/Arbutus Ridge/Kerrisdale is more senior-driven, with the highest senior share (19.81%). Overall priority differs: Sunset ranks first overall despite placing fifth on this proxy, so population size and the other priority components remain important planning context.

Methodological caution

Age structure is used here as a vulnerability proxy. It does not measure socioeconomic disadvantage, disease burden, healthcare utilization, unmet need, or validated clinical demand.

Data Blindspots

Where census uncertainty is highest across Vancouver communities

General Non-Response Rate is treated as a data-quality indicator, not a measure of socioeconomic vulnerability or healthcare need.

CHSA Name

CHSA Name

All



Total Population

663.9K

Weighted GNR

6.51%

Data Blindspot (Avg./Selected)

52.63

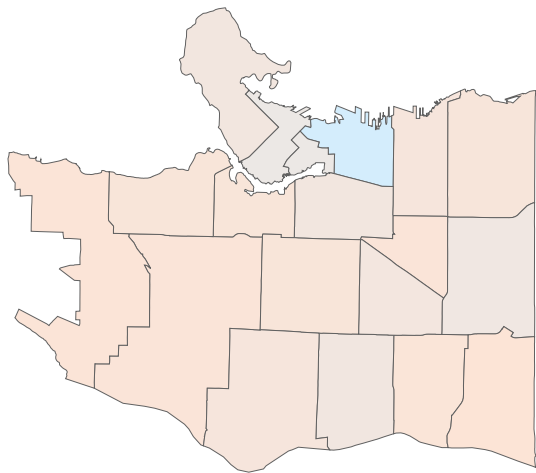
Blindspot Contribution (Avg./Selected)

13.16

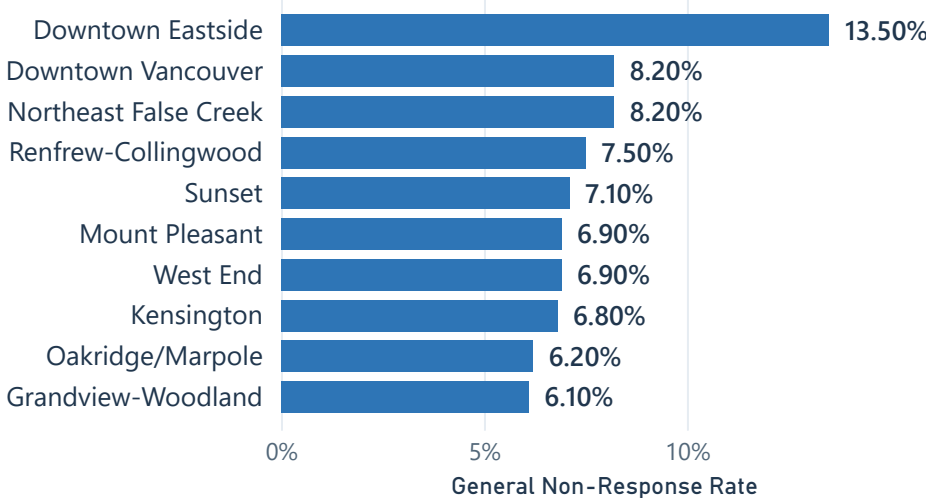
Total CHSAs

19

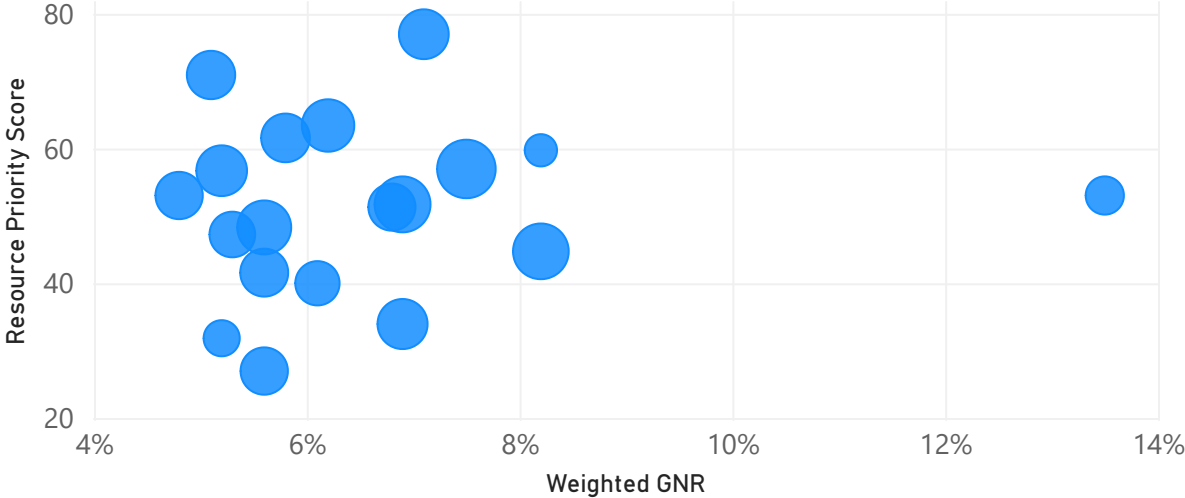
Where Census Uncertainty Is Highest



Communities with the Highest General Non-Response Rate

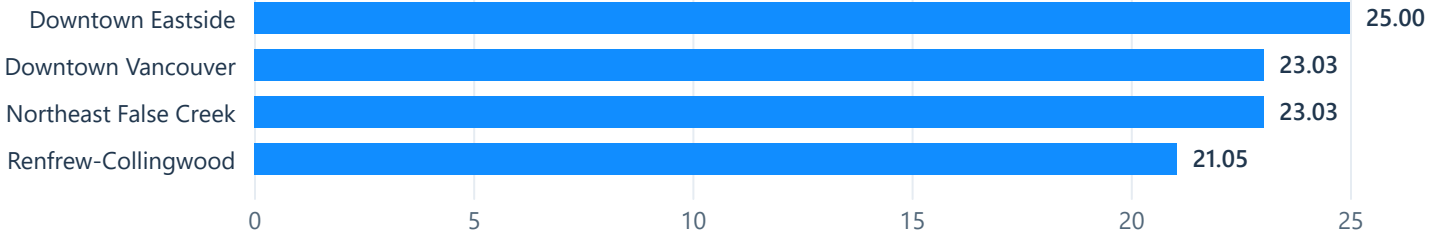


Census Uncertainty vs. Overall Resource Priority



Data Uncertainty Contribution to Resource Priority

Balanced contribution: 25% × Data Blindspot Score.



Data Blindspot Evidence

CHSA Name	Weighted GNR	Data Blindspot Score	Blindspot Priority Contribution	Resource Priority Score	Priority Rank
Downtown Eastside	13.50%	100.00	25.00	53.03	8
Downtown Vancouver	8.20%	92.11	23.03	44.74	14
Northeast False Creek	8.20%	92.11	23.03	59.74	5
Renfrew-Collingwood	7.50%	84.21	21.05	56.97	6
Total	6.51%	52.63	13.16	51.05	1

Management Insight

Downtown Eastside has the highest Long-Form GNR at 13.5%, indicating the greatest census-response uncertainty in the Vancouver-focused dataset. Northeast False Creek and Downtown Vancouver both have Data Blindspot Scores of 92.11, yet rank 5th and 14th overall respectively. Census uncertainty should inform confidence in planning evidence, not be interpreted as vulnerability or healthcare need.

Methodological caution

General Non-Response Rate is a census data-quality indicator. Higher values indicate greater uncertainty in the underlying census evidence; they do not indicate socioeconomic disadvantage, healthcare need, utilization, unmet need, or service demand.

Healthcare Supply & Service Gaps

Where mapped healthcare facility locations are relatively scarce across Vancouver communities

ODHF facility counts are used as a supply-location proxy and do not represent healthcare capacity, staffing, utilization or unmet need.

CHSA Name

CHSA Name

All



Total Population

663.9K

Total Facilities

153

Facilities per 10K Population

2.30

Supply Gap Score (Avg./Selected)

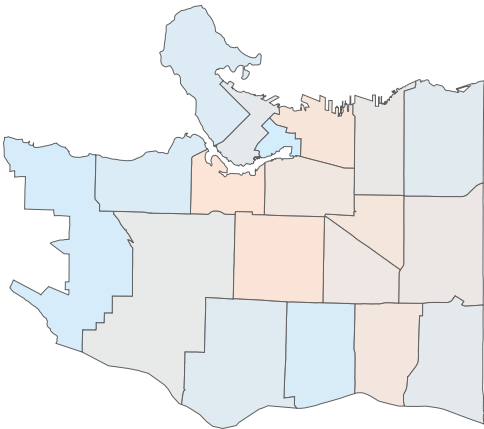
47.37

Total CHSAs

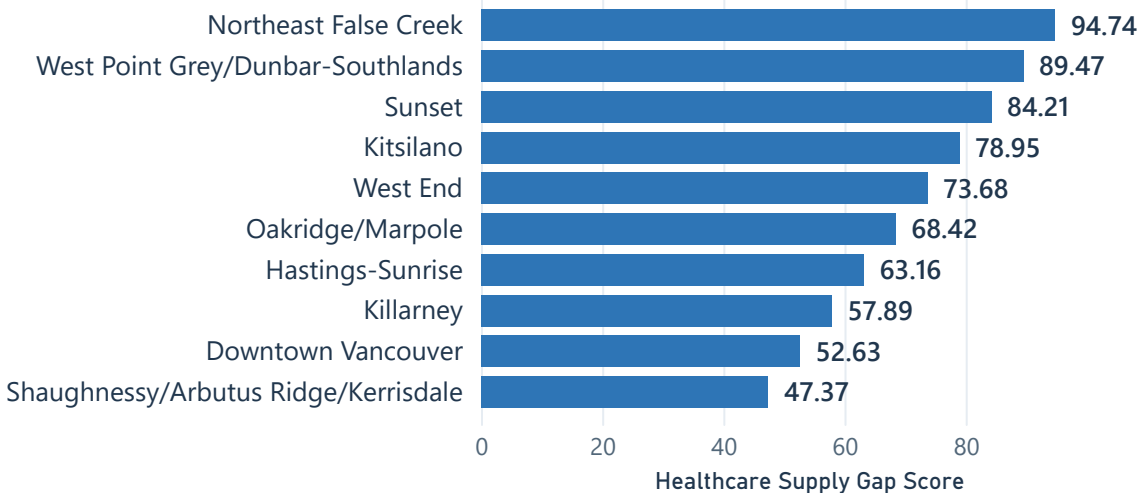
19

Where Mapped Facility Supply Gaps Are Largest

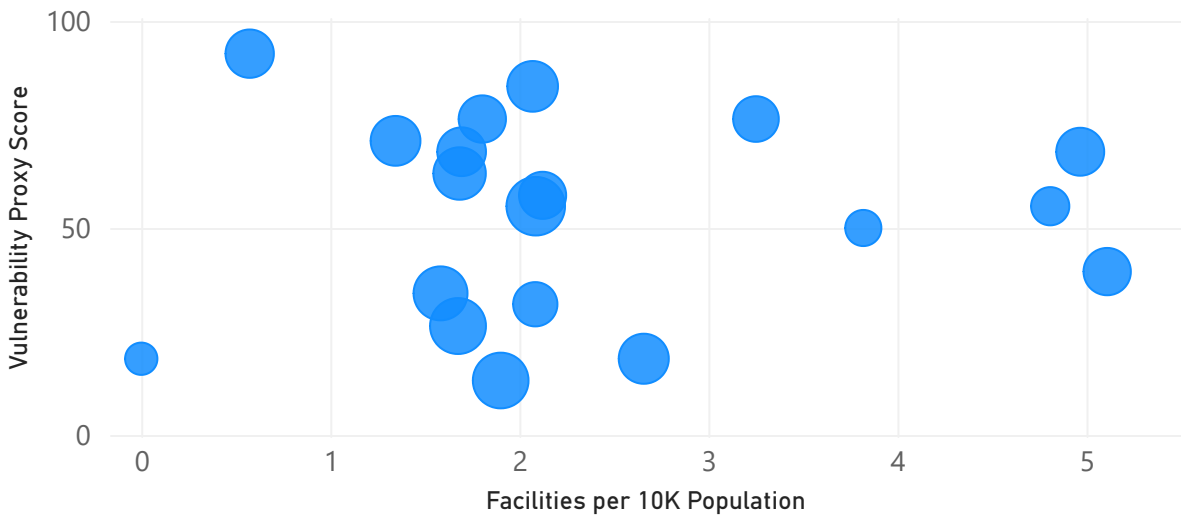
Higher score = fewer mapped facility locations relative to population.



Communities with the Largest Mapped Supply Gaps



Mapped Supply Gap vs. Age-Structure Vulnerability



Mapped Healthcare Facilities Across Vancouver

Each point is a mapped facility location; point size does not represent capacity.

Healthcare Supply & Gap Detail

CHSA Name	Total Population	Total Facilities	Facilities per 10K Population	Healthcare Supply Gap Score	Vulnerability Proxy Score
Northeast False Creek	14223.0	0	0.00	94.74	1
West Point Grey/Dunbar-Southlands	34866.0	2	0.57	89.47	9
Sunset	37170.0	5	1.35	84.21	7
Kitsilano	37170.0	5	1.35	78.95	7
West End	37170.0	5	1.35	73.68	7
Oakridge/Marpole	37170.0	5	1.35	68.42	7
Hastings-Sunrise	37170.0	5	1.35	63.16	7
Killarney	37170.0	5	1.35	57.89	7
Downtown Vancouver	37170.0	5	1.35	52.63	7
Shaughnessy/Arbutus Ridge/Kerrisdale	37170.0	5	1.35	47.37	7
Total	663894.0	153	2.30	47.37	5.0

Management Insight

Fairview has the highest mapped facility rate at 5.11 per 10,000 residents and the lowest modeled supply-gap score. West Point Grey/Dunbar-Southlands has 0.57 mapped facilities per 10,000, the second-highest modeled supply gap, and the highest age-structure vulnerability proxy. Northeast False Creek has zero mapped ODHF facilities; these are investigation signals, not proof of unmet need or limited access.

Methodological caution

Mapped facility locations are a supply-location proxy, not a measure of healthcare capacity, staffing, service volume, utilization, accessibility, ownership, quality, outcomes or unmet need. Nearby cross-boundary services and facilities without usable coordinates may also affect interpretation

Resource Allocation Scenario

How a hypothetical resource pool changes when community priority complements population share

SCENARIO MODEL — NOT ACTUAL VCH RESOURCE ALLOCATION

Hypothetical allocation uses 60% population share and 40% modeled resource priority. Values are planning-scenario outputs, not actual budgets, staffing decisions or service commitments.

CHSA Name

CHSA Name

All



Scenario Budget Allocation

\$10.00M

Scenario Clinical FTE Allocation

50.00

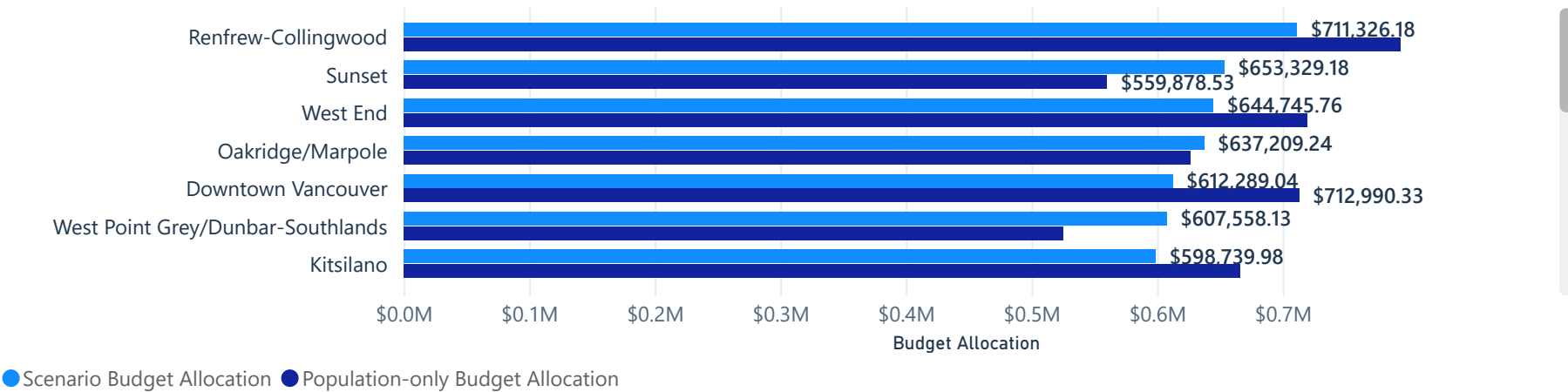
Scenario Community Care Allocation

1000

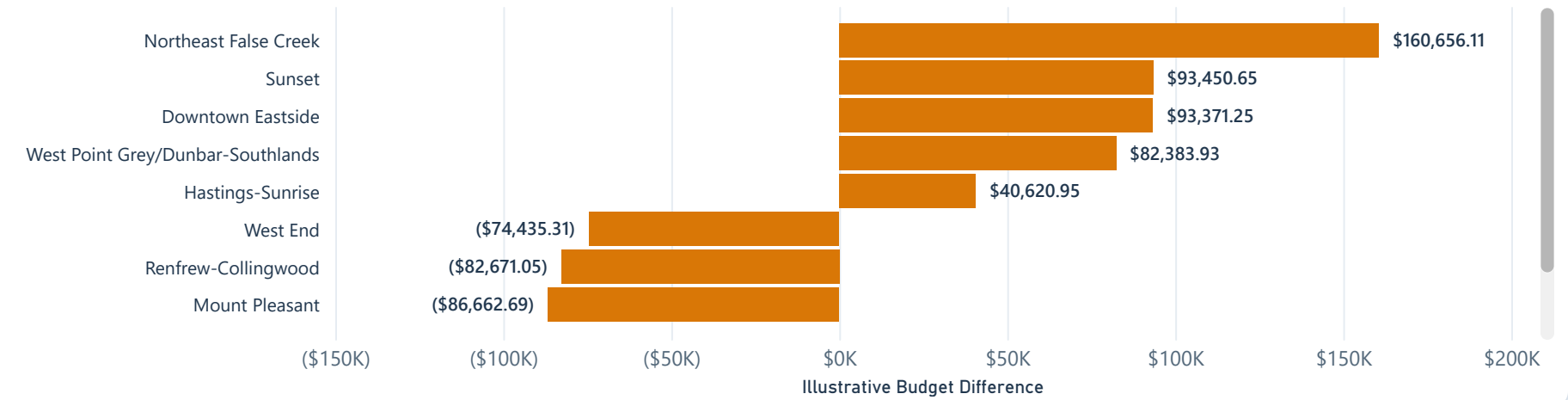
High Priority Communities

5

Hypothetical Budget Allocation by Community



Budget Shift vs. Population-Only Baseline



Hypothetical Resource Allocation Detail

CHSA Name	Scenario Budget Allocation	Budget Difference	Scenario Clinical FTE Allocation	Scenario Community Care Allocation	Resource Priority Score	Priority Rank
Renfrew-Collingwood	\$711,326.18	(\$82,671.05)	3.50	71	56.97	1
Sunset	\$653,329.18	\$93,450.65	3.30	65	76.97	2
West End	\$644,745.76	(\$74,435.31)	3.20	64	51.71	10
Oakridge/Marpole	\$637,209.24	\$11,070.13	3.20	64	63.42	7
Downtown Vancouver	\$612,289.04	(\$100,701.29)	3.10	61	44.74	15
Total	\$10,000,000.00	\$0.00	50.00	1000	51.05	-

Priority Rank Sensitivity

CHSA Name	Balanced Rank	Need-focused Rank	Uncertainty-aware Rank	Supply-gap-focused Rank
Fairview	19	18	19	19
Cedar Cottage	18	16	18	18
Mount Pleasant	17	19	16	16
Grandview-Woodland	16	15	15	15
South Cambie/Riley Park	15	13	17	17
Total	1	1	1	1

Management Insight

The priority-informed scenario changes allocation relative to population alone because it also incorporates age-structure vulnerability, census uncertainty and relative mapped facility-location scarcity. Northeast False Creek shows the largest positive illustrative budget shift (+\$160,656.11), while Sunset also receives a larger modeled share and remains highly ranked across weighting scenarios. These shifts identify communities for further planning investigation; they do not establish actual funding need or recommended VCH allocations. Scenario totals reconcile exactly to \$10M, 50 FTE and 1,000 community-care units.

Scenario caution

Allocation results depend on explicit modeling assumptions. Priority weights change rankings, and the public datasets do not measure facility capacity, staffing, utilization, wait times, patient flows, accessibility, outcomes or current clinical need. Operational decisions require additional data and stakeholder validation.

Methodology & Data Quality

How public data, transparent scoring and validation support the planning scenario

Independent analytical model built from public/open data. Scores are relative planning proxies and scenario outputs, not measures of clinical need or actual VCH resource decisions.

Passed Controls

13

Failed Controls

0

Total Facilities

153

Total CHSAs

19

Analytical Pipeline

Public Data → Geographic Validation → Community Profile → Vulnerability Proxy → Data Blindspot → Healthcare Supply Gap → Resource Priority → Scenario Allocation

Public census profiles provide population and age structure. Census non-response remains a separate uncertainty signal. Public facility locations are spatially assigned to CHSAs before the three components are combined in a transparent planning scenario.

Public Data Sources

Statistics Canada — 2016 Census / Census Profile

Population, age structure and long-form census non-response.

Open Database of Healthcare Facilities (ODHF)

Facility names, types and coordinates for public facility-location mapping. Locations are not capacity, staffing, utilization, accessibility, quality, outcomes or VCH ownership.

BC CHSA geography

Community-level analytical units.

Vancouver Census Subdivision boundary

City overlap and spatial-scope validation.

What the Scores Mean

Vulnerability Proxy Score

Relative age-structure proxy. Not clinical need, deprivation or a validated multidimensional vulnerability index.

Data Blindspot Score

Relative census non-response signal. Higher values mean greater data uncertainty—not disadvantage, clinical vulnerability or poor access.

Healthcare Supply Gap Score

Inverse signal from mapped facility locations per resident. Higher values mean fewer mapped locations relative to population—not capacity, quality, access or unmet need.

Resource Priority Methodology

45% Vulnerability Proxy + 25% Data Blindspot + 30% Healthcare Supply Gap = Resource Priority Score

Component contributions remain visible so users can see why a community ranks highly. These weights are explicit assumptions, not validated allocation coefficients.

Sensitivity Testing

Balanced 45 / 25 / 30 · Need-focused 60 / 10 / 30

Uncertainty-aware 35 / 35 / 30 · Supply-gap-focused 35 / 15 / 50

(Vulnerability / Blindspot / Supply Gap)

Sunset, West Point Grey/Dunbar-Southlands and Oakridge/Marpole remain Top 5 in all four scenarios; Sunset ranks first or second throughout. Rank movement confirms dependence on assumptions—not actual need.

Hypothetical Allocation Logic

60% Population Share + 40% Resource Priority Share

Applied to hypothetical pools:

- \$10,000,000 budget
- 50 clinical FTE
- 1,000 community-care units

Population-only allocation is retained as the baseline. Exact largest-remainder reconciliation preserves all three pool totals.

These are scenario outputs—not actual VCH allocations, commitments or recommendations.

Automated Validation Results

13 controls passed · 0 failed

- ✓ 19 Vancouver-focused CHSAs; unique keys
- ✓ UBC 3243 excluded; partial boundary flagged
- ✓ 153 mapped facilities; unique keys
- ✓ Referential integrity passed
- ✓ 0 null retained spatial assignments
- ✓ All scores within 0–100
- ✓ \$10M, 50 FTE and 1,000 units reconcile
- ✓ 76 sensitivity rows complete
- ✓ Population total stable at 663,894

Source: ValidationResults[Test, Status, Detail]

Geographic & Data Quality Notes

- Scope contains 19 Vancouver-focused CHSAs.
- UBC CHSA 3243 is excluded by the Vancouver City overlap rule.
- One retained facility required approximately 25 m edge tolerance.
- No retained mapped facility has a null CHSA assignment.
- 29 Vancouver candidate facility records lacked coordinates and may undercount mapped supply.
- Cross-boundary services are not modeled as patient-access flows.
- Zero mapped facilities does not prove absence of healthcare access.

Key Limitations

- Census profile uses 2016 data.
- Age structure is a proxy, not socioeconomic or clinical need.
- GNR measures uncertainty, not disadvantage.
- ODHF locations do not measure capacity, staffing, hours, ownership, utilization, access, quality or outcomes.
- Missing coordinates and unmodeled cross-boundary flows affect interpretation.
- Rankings depend on assumed weights.
- Allocations are hypothetical—not VCH resources or recommendations.
- The model does not establish causal demand or unmet need.

Before operational use: add capacity, staffing, utilization, waits, patient flows, accessibility, service scope, outcomes, current population data and stakeholder/clinical validation.

Independent portfolio project · Public/open data · Scenario model, not actual VCH allocation